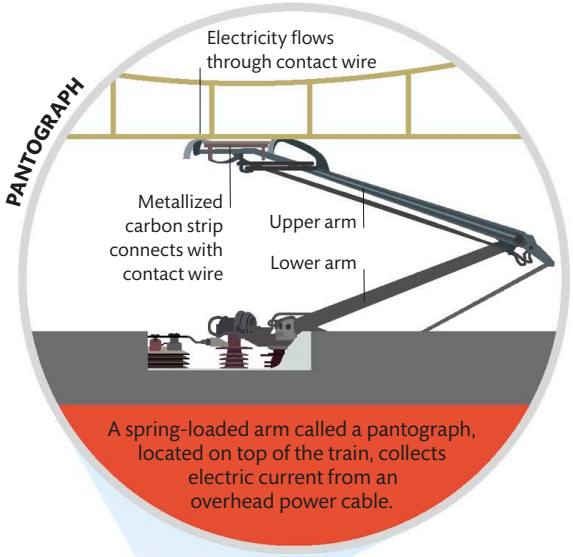
Trains

Trains provide one of the most time-efficient transportation solutions for traveling long distances. Most modern trains are powered by a diesel engine or an external source of electricity.

Electric trains

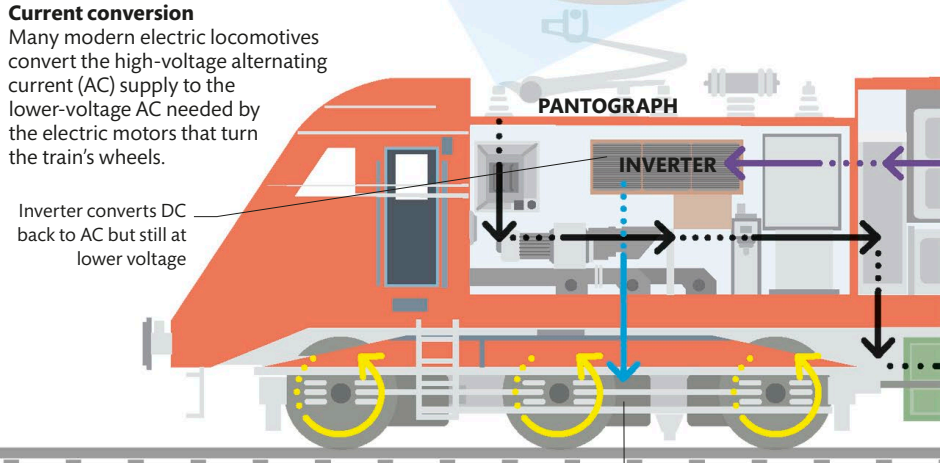
Electric trains are powered by electricity supplied either from overhead cables or by a third rail in the track. Since they do not have to carry their own power-generating equipment, electric locomotives are lighter than diesel equivalents and are therefore capable of faster acceleration.



WHO BUILT THE FIRST RAILROAD LOCOMOTIVE?

In 1804, English engineer Richard Trevithick built the first railroad locomotive. It was used to haul iron from Penydarren Ironworks in Wales.

Current conversion
Many modern electric locomotives convert the high-voltage alternating current (AC) supply to the lower-voltage AC needed by the electric motors that turn the train's wheels.



KEY
High-voltage AC · Lower-voltage DC · Lower-voltage AC · Fuel

Diesel-electric trains

Most modern diesel trains employ a diesel-electric powerplant housed inside the locomotive. Rather than power the wheels directly, the diesel engine drives a generator or alternator (see pp.16–17) to produce electricity, which operates the train's electrical systems and traction motors. Since diesel trains need no external power supply, they are used on rail lines where electrification is uneconomical.

Engine power
The alternating current (AC) from the engine-driven alternator is converted to direct current (DC) by a rectifier. An inverter converts this into AC to supply the motors.

